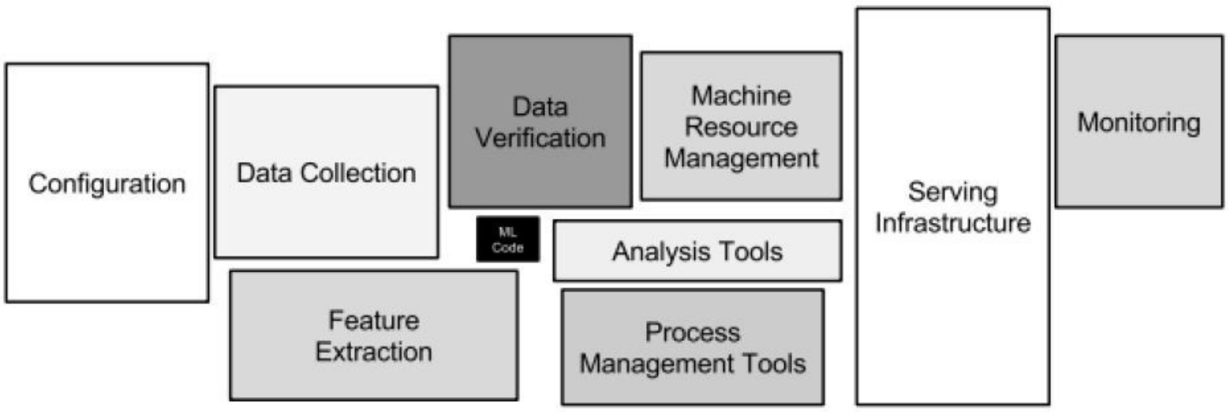
Cloud computing

ONE LOVE. ONE FUTURE.

1

Technical depth in machine learning

A fraction of real world ML systems is composed of ML code, the rest is infrastructure.



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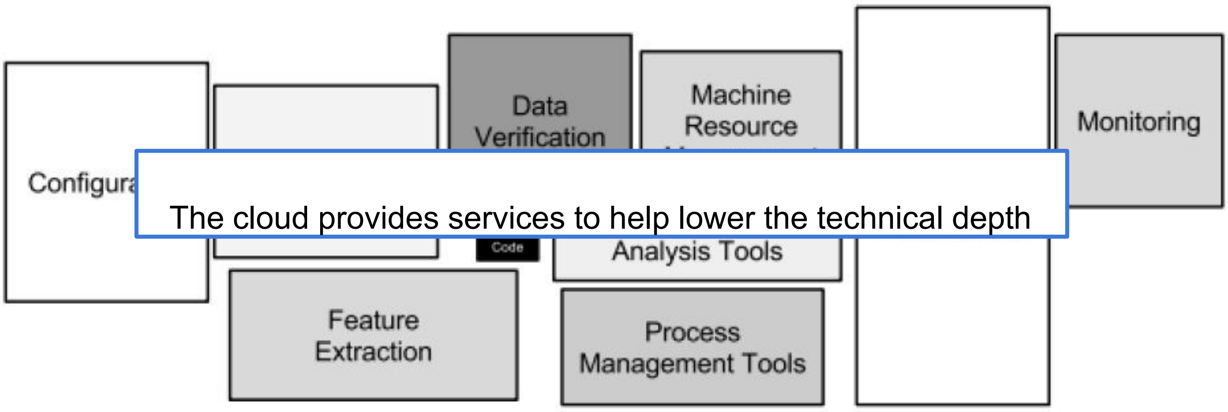

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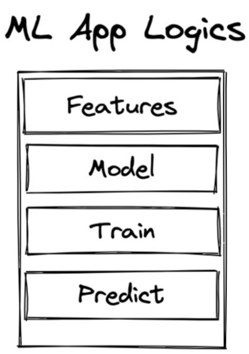
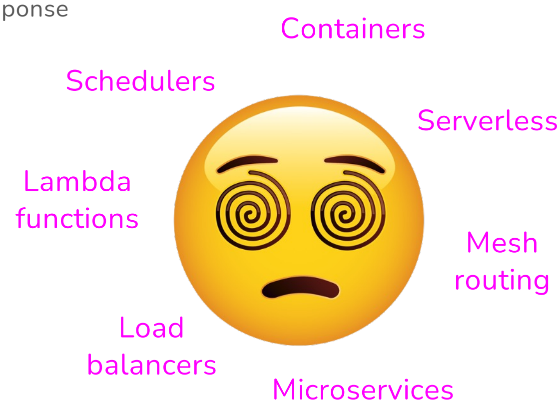
ML systems are complex

- More components
  - A request might jump 20-30 hops before response
  - A problem occurs, but where?





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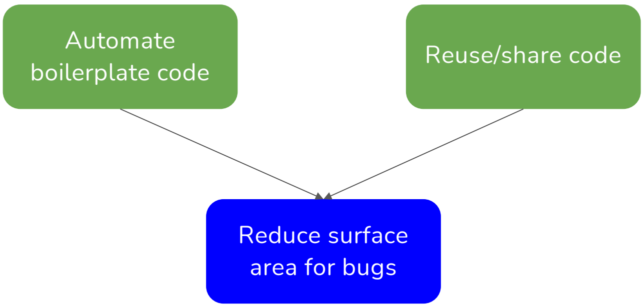
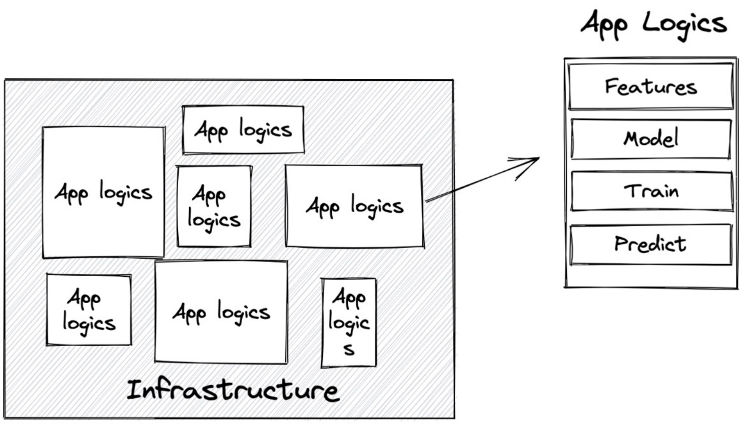
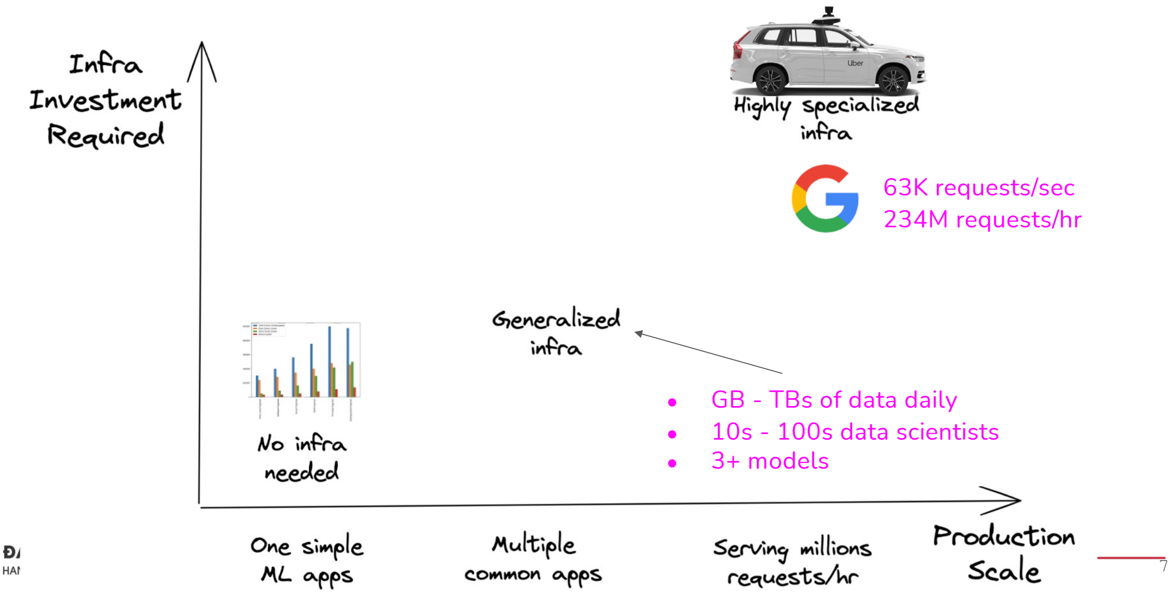
More complex systems, better infrastructure needed



- Infrastructure: the set of fundamental facilities and systems that support the sustainable functionality of households and firms.
- ML infrastructure: the set of fundamental facilities that support the development and maintenance of ML systems.



Every company's infrastructure needs are different



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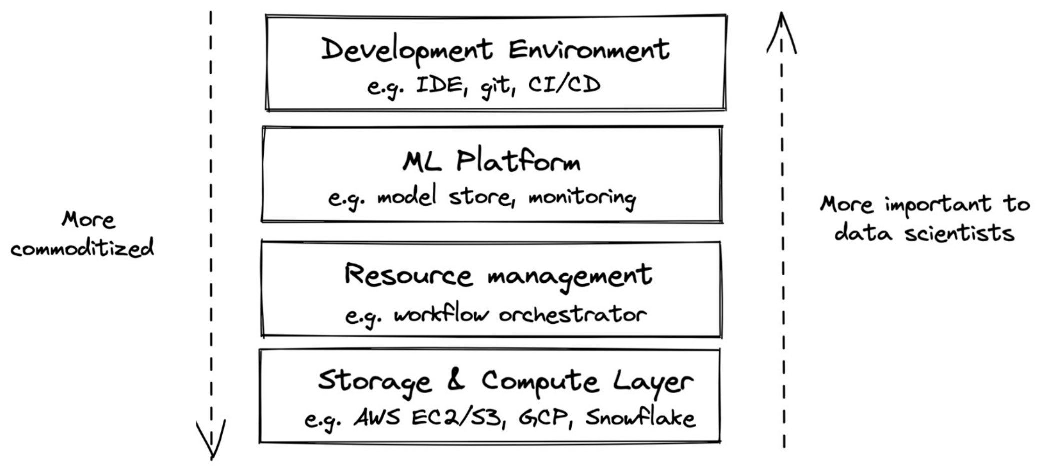
Infrastructure Layers



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Storage & Compute Layer

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Storage

- Where data is collected and stored
- Simplest form: HDD, SSD
- More complex forms: data lake, data warehouse
- Examples: S3, Redshift, Snowflake, BigQuery

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Storage: heavily commoditized

- Most companies use storage provided by other companies (e.g. cloud)
- Storage has become so cheap that most companies just store everything

Compute layer: engine to execute your jobs

- Compute resources a company has access to
- Mechanism to determine how these resources can be used

Compute layer: engine to execute jobs

- Simplest form: a single CPU/GPU core
- Most common form: cloud compute

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Compute unit

- Compute layer can be sliced into smaller compute units **to be used concurrently**
  - A CPU core might support 2 concurrent threads, **each thread** is used as a compute unit to execute its own **job**
  - Multiple CPUs can be joined to form a **large compute unit** to execute a large **job**

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Compute unit

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Unit: job | **Unit: pod** ← Wrapper around container

DASK

APACHE Spark

RAY

kubernetes

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Compute layer: how to execute jobs

1. Load data into memory
2. Perform operations on that data
   a. Operations: add, subtract, multiply, convolution, etc.

To add arrays A and B
1. Load A & B into memory
2. Perform addition on A and B

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Compute layer: how to execute jobs

1. Load data into memory
2. Perform operations on that data
   a. Operations: add, subtract, multiply, convolution, etc.

If A & B don't fit into memory, it'll be possible to do the ops without out-of-memory algorithms

To add arrays A and B
1. Load A & B into memory
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Compute layer: how to execute jobs

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Important metrics of compute layer:
1. Memory
2. Speed of computing ops

Compute layer: memory

- Amount of memory
  - Straightforward
  - An instance with 8GB of memory is more expensive than an instance with 2GB of memory

Compute layer: memory

- Amount of memory
- I/O bandwidth: speed at which data can be loaded into memory

Compute layer: speed of ops

- Most common metric: FLOPS
  - **F**loating **P**oint **O**perations **P**er **S**econd

"A Cloud TPU v2 can perform up to 180 teraflops, and the TPU v3 up to 420 teraflops."

- Google, 2021

Compute layer: speed of ops

- Most common metric: FLOPS
- Contentious
  - What exactly is an ops?
    - If 2 ops are fused together, is it 1 or 2 ops?
  - Peak perf at 1 teraFLOPS doesn't mean your app will run at 1 teraFLOPS

Compute layer: utilization

- Utilization = actual FLOPS / peak FLOPS

If peak 1 trillion FLOPS but job runs 300 billion FLOPS
-> utilization = 0.3

Compute layer: utilization

- Utilization = actual FLOPS / peak FLOPS
- Dependent on how fast data can be loaded into memory

The higher, the better

Tensor Cores are very fast. So fast … that they are idle most of the time as ***they are waiting for memory to arrive from global memory.***

For example, during BERT Large training, which uses huge matrices — the larger, the better for Tensor Cores — ***we have utilization of about 30%****.*

\- Tim Dettmers, 2020

Compute layer: evaluation

- Memory
- Cores
- I/O bandwidth
- Cost

Some GPU instances on AWS

Instance	GPUs	vCPU	Mem (GiB)	GPU Mem (GiB)
p3.2xlarge	1	8	61	16
p3.8xlarge	4	32	244	64
p3.16xlarge	8	64	488	128
p3dn.24xlarge	8	96	768	256

Some TPU instances on GCP

TPU type (v2)	v2 cores	Total memory
v2-8	8	64 GiB
TPU type (v3)	v3 cores	Total memory
v3-8	8	128 GiB

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Conceptually: What is the cloud

It is to data scientists what advance calculus was to mathematicians in the early 17th century:

The power of infinity

"Cloud computing", Stable diffusion v1

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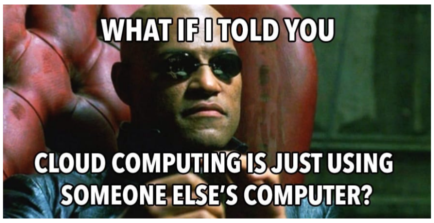
Cloud Computing Overview

- Running local computations is sufficient for initial code development.
- Scaling experiments requires more computing power than a standard laptop/desktop.
- Cloud computing as a solution for scaling.
- Assumes prior experience with local clusters or similar.



https://towardsdatascience.com/how-to-start-a-data-science-project-using-google-cloud-platform-6618b7c6edd2

In practise

1. Big data centers of interconnected computers.
2. A software stack on top

Image credit: https://www.nytimes.com/2020/02/27/technology/cloud-computing-energy-usage.html

The advantage/disadvantage of cloud

Pros:

- Extensive collection of services
- Infinite scaling
- Documentation and help

Cons:

- It's expensive over longer time compared to alternatives
- It can be hard to get started
- Different vendors do things differently

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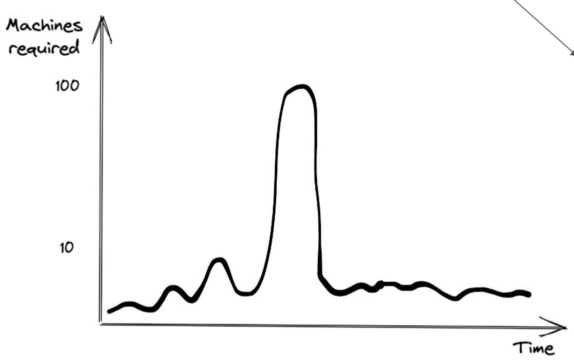
Benefits of cloud

- Easy to get started
- Appealing to variable-sized workloads
  - Private: would need 100 machines upfront, most will be idle most of the time
  - Cloud: pay for 100 machines only when needed

Autoscaling!



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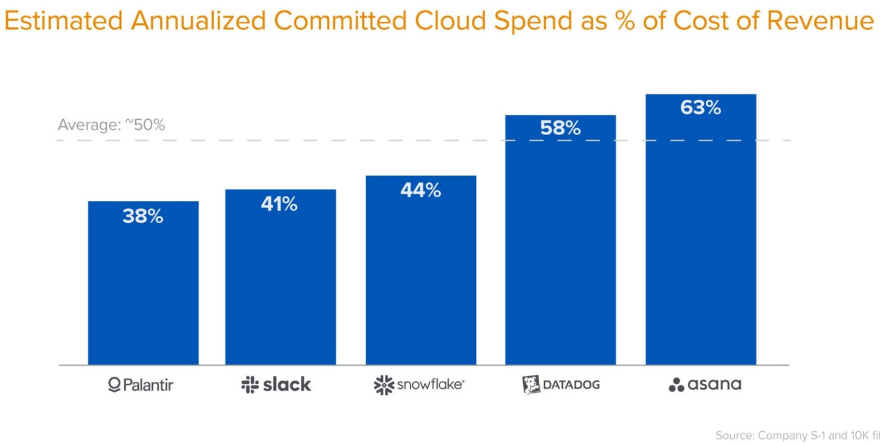
Drawbacks of cloud: cost

- Cloud spending: ~50% cost of revenue



The Cost of Cloud, a Trillion Dollar Paradox | Andreessen Horowitz (2021)

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Drawbacks of cloud: cost

"*Across 50 of the top public software companies currently utilizing cloud infrastructure, an **estimated $100B of market value is being lost … due to cloud impact on margins** — relative to running the infrastructure themselves.*"

The Cost of Cloud, a Trillion Dollar Paradox | Andreessen Horowitz (2021)

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Cloud vendors

AWS are still the largest player in the cloud market.

However the others are growing faster.

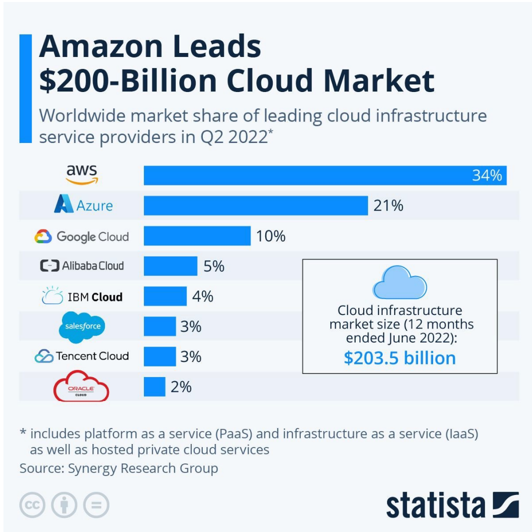

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Image credit: https://uk.pcmag.com/old-cloud-infrastructure/131713/four-companies-control-67-of-the-worlds-cloud-infrastructure

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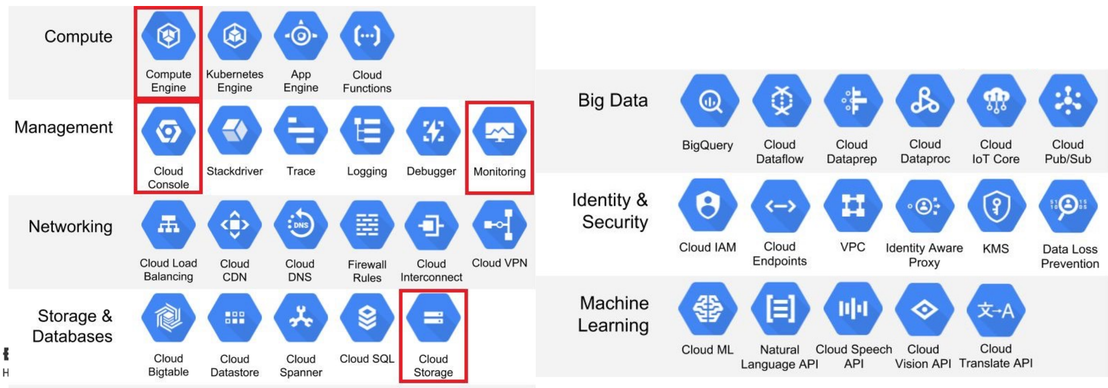
What does the cloud consist of

Each cloud vendor have a number of services

Depending on your application, only a subset is of interest



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The important services for ML

Engine	General compute
Storage	General storage
Functions/Run	Deployment
Vertex	Training of AI

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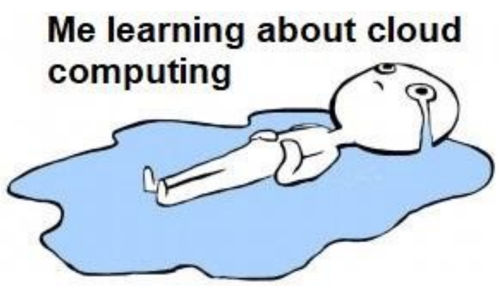
A word of warning

Working in the cloud is…hard.

- Everything taking longer because extra layer of communication
- Syntax can be hard to remember
- A lot of services to be confused about

The only way to learn is to use it.

If you can, start simple and then scale up



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Cloud computing providers

There exist numerous amount of cloud computing providers with some of the biggest being:

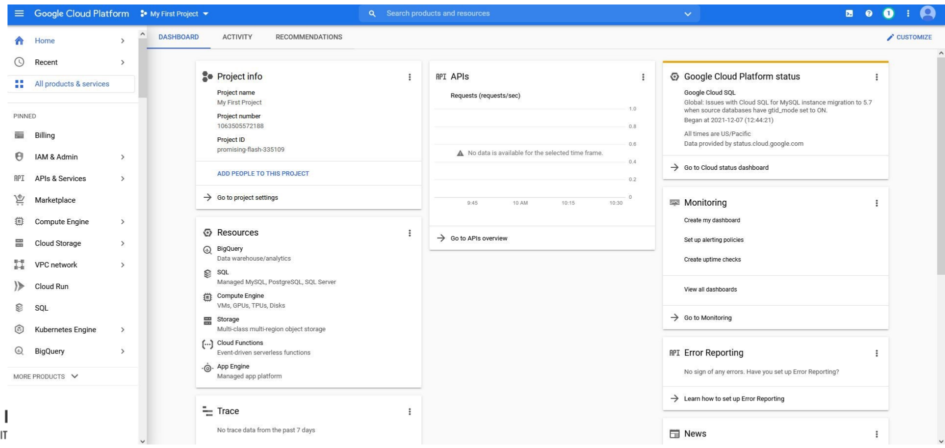
- Azure
- AWS
- Google Cloud project
- Alibaba Cloud

Google Cloud

- Slight advantages/disadvantages among cloud providers.
- Google Cloud
  - Free credit available when signing up with a new account.
- Key point: Most cloud providers offer the same set of services.
  - Learning one provider often translates to understanding others.
  - Differences mainly in service names and interfaces, but functionality is similar.

Lets take a look

https://console.cloud.google.com



Cloud Exercises

- Focus: Getting familiar with working on the cloud.
- For doubts or deeper exploration:
  - Recommended: series of videos.
  - Resource: General documentation on Google Cloud.

The learning objectives:

- In general being familiar with the Google SDK working
- Being able to start different compute instances and work with them
- Know how to do continuous integration workflows for the building of docker images
- Knowledge about how to store data and containers/artifacts in cloud buckets
- Being able to train simple deep-learning models using a combination of cloud services

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Cloud setup

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Google Cloud Project (GCP)

- GCP is Google's cloud service platform
- A key selling point of cloud providers is offering near-infinite resources
- Cloud enables scaling for deep learning and machine learning tasks
- Local resources are often insufficient for modern AI tasks

https://www.pintonista.com/google-cloud-platform-intro/

? Exercises

1. Login to the GCP Homepage
2. Check Cloud Credit in Billing
3. Create a New Project in GCP
4. Install and Configure `gcloud` CLI
5. Enable Google Cloud Developer APIs

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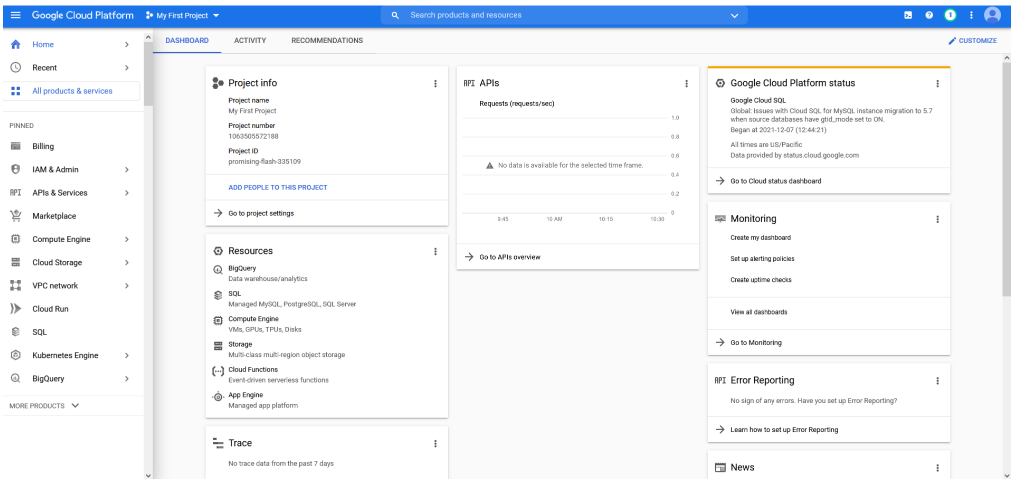
Login to the GCP Homepage



Image

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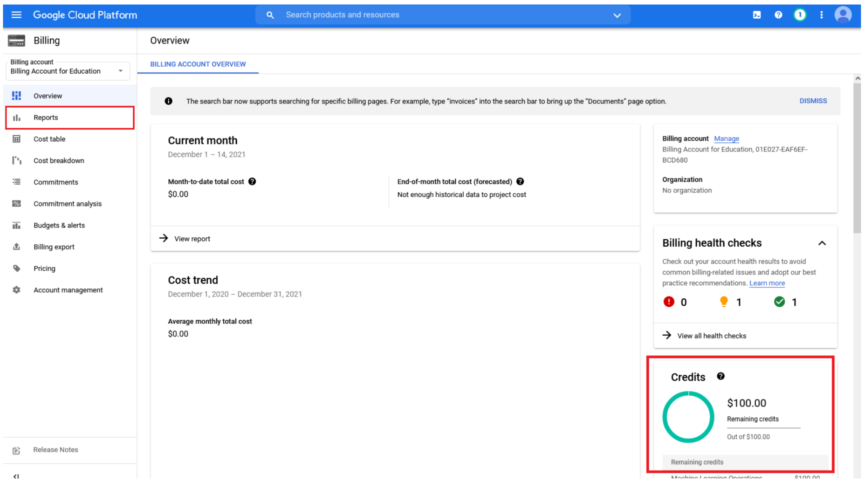
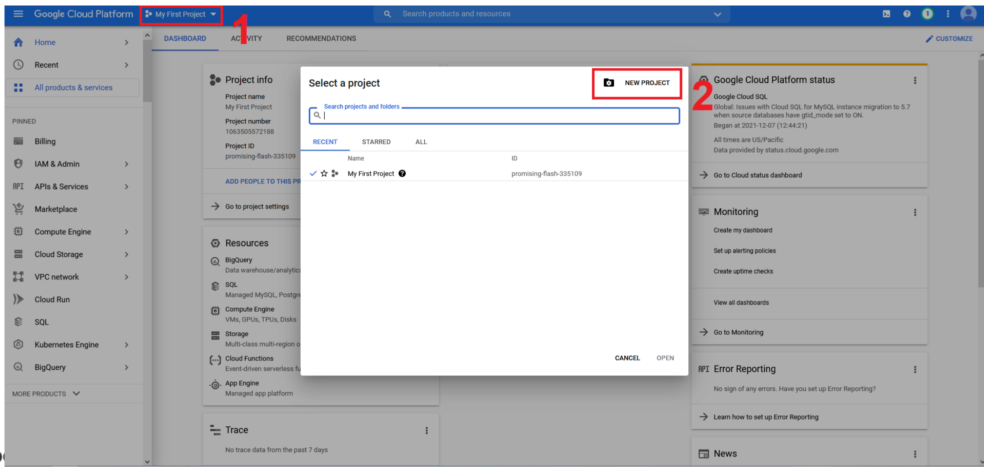
Check Cloud Credit in Billing



Image

Create a New Project in GCP

- One way to stay organized within GCP is to create projects.
- Create a new project called `mlops`. When you click `create` you should get a notification that the project is being created.



Install and Configure `gcloud` CLI

- Install `gcloud`, the command-line interface for managing Google Cloud accounts
- `gcloud` can perform nearly all tasks available through the web interface
- Follow the installation instructions specific to your OS here

1. After installation, try in a terminal to type:

```
gcloud -h
```

the command should and show the help page. If not, something went wrong in the installation (you may need to restart after installing).

2. Now login by typing
 `gcloud auth login`
 you should be sent to an web page where you link your cloud account to the `gcloud` interface. Afterwards, also run this command:
 `gcloud auth application-default login`
 If you at some point want to revoke this you can type:
 `gcloud auth revoke`

3. Next you will need to set the project that we just created. In your web browser under project info, you should be able to see the `Project ID` belonging to your `dtumlops` project. Copy this an type the following command in a terminal
 `gcloud config set project <project-id>`
 You can also get the project info by running
 `gcloud projects list`

4. Next install the Google Cloud Python API:
```
pip install --upgrade google-api-python-client
```
Make sure that the Python interface is also installed. In a Python terminal type
```
import googleapiclient
```
this should work without any errors.

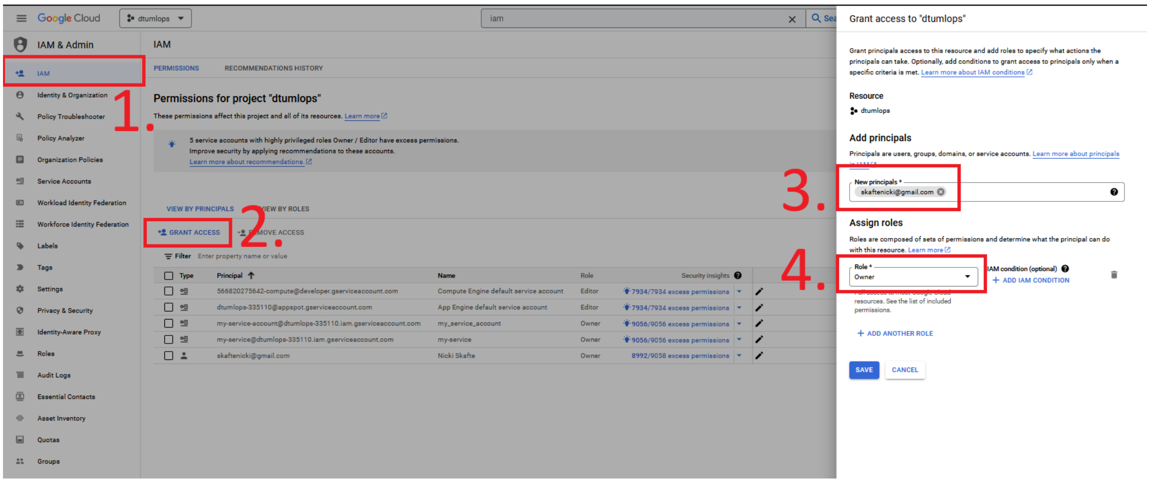
5. (Optional) If you are using VSCode you can also download the relevant extension called `Cloud Code`. After installing it you should see a small `Cloud Code` button in the action bar.

Enable Google Cloud Developer APIs

- Activate necessary APIs not enabled by default:
  - API Gateway: `gcloud services enable apigateway.googleapis.com`
  - Service Management: `gcloud services enable servicemanagement.googleapis.com`
  - Service Control: `gcloud services enable servicecontrol.googleapis.com`
- Check enabled services with: `gcloud services list`

IAM and Quotas

- Admin involves managing roles and resource quotas for GCP users
- Quotas limit access to resources, such as GPUs, to control costs
  - Example: Data scientists may have limited GPU access, while DevOps engineers may need different services
- Focus is not on this aspect in the course, but it's important to understand
- To share a project with others:
  - Use the IAM (Identity and Access Management) page
  - Click the `Grant Access` button
  - Search for the person's email and assign `Viewer`, `Editor`, or `Owner` access depending on the needed permissions



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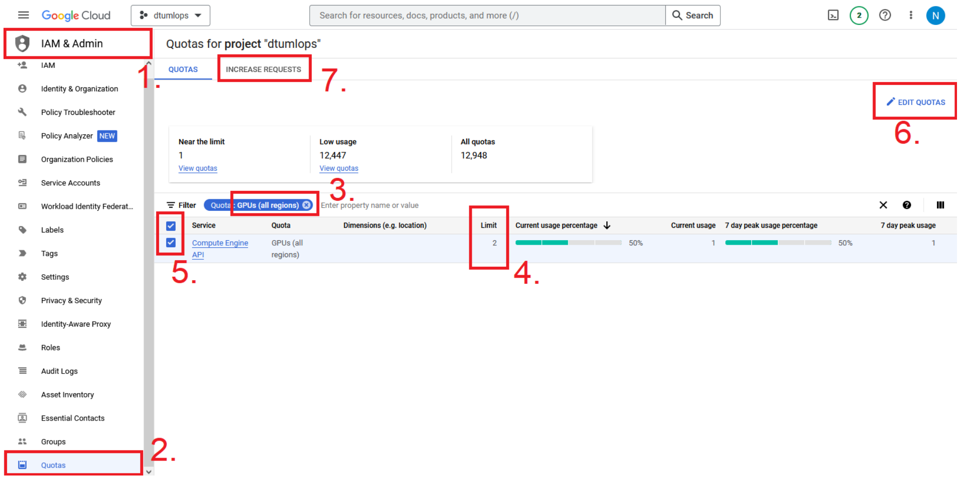
Exercises: Enabling Compute Engine and Managing GPU Quotas

1. **Enable Compute Engine:**
   - Search for “Compute Engine” in the top search bar
   - Enable the service (this may take some time)
2. **Adjust GPU Quotas:**
   - Go to the `IAM & Admin` page via the search bar
   - Follow these steps to manage GPU quotas:
     1. Go to the `Quotas` page
     2. Search for `GPUs (all regions)` (exact match, case-sensitive)
     3. Check current GPU quota and usage
     4. Click the quota and select `Edit` quotas
     5. In the pop-up, increase the limit to 1 or 2 GPUs
     6. Check request status under the `Increase requests` tab



Image

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Handling Quota Errors and Increase Requests

- If you encounter GPU-related errors mentioning `quotas`, check your

Filter: Service : Vertex AI API | Metric : aiplatform.googleapis.com/custom_model_training_nvidia_k80_gpus | Enter property name or value

Service	Quota	Dimensions (e.g. location)	Limit	Current usage percentage	Current usage
Vertex AI API	Custom model training Nvidia K80 GPUs per region (default)		0		
Vertex AI API	Custom model training Nvidia K80 GPUs per region	region : asia-east1	0	0%	0
Vertex AI API	Custom model training Nvidia K80 GPUs per region	region : asia-east2	0	0%	0
Vertex AI API	Custom model training Nvidia K80 GPUs per region	region : asia-northeast1	0	0%	0
Vertex AI API	Custom model training Nvidia K80 GPUs per region	region : asia-northeast2	0	0%	0
Vertex AI API	Custom model training Nvidia K80 GPUs per region	region : asia-northeast3	0	0%	0

mining bots)

Image

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🧠 Knowledge check

1. What considerations to take when choosing an GCP region for running a new application?
   - Services availability in the region, not all services are available in all regions
   - Resource availability: some regions have more GPUs available than others
   - Reduced latency: if your application is running in the same region as your users, the latency will be lower
   - Compliance: some countries have strict rules that require user info to be stored inside a particular region eg. EU has GDPR rules that require all user data to be stored in the EU
   - Pricing: some regions may have different pricing than others

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🧠 Knowledge check

2. The 3 major cloud providers all have the same services, but they are called something different depending on the provider. What are the corresponding names of these GCP services in AWS and Azure?
   - Compute Engine
   - Cloud storage
   - Cloud functions
   - Cloud run
   - Cloud build
   - Vertex AI

   It is important to know these correspondences to navigate blogpost etc. about MLOps on the internet.

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Solution

GCP	AWS	Azure
Compute Engine	Elastic Compute Cloud (EC2)	Virtual Machines
Cloud storage	Simple Storage Service (S3)	Blob Storage
Cloud functions	Lambda	Functions Serverless Compute
Cloud run	App Runner, Fargate, Lambda	Container Apps, Container Instances
Cloud build	CodeBuild	DevOps
Vertex AI	SageMaker	AI Platform